

David Brooks

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EDUCATION

University of California, San Diego

Fall 2022 - Present

Mechanical Engineering Undergraduate with Specialization in Fluid Mechanics and Thermal Systems

♦ Provost's Honors - Winter 2023 | 3.65 GPA

SKILLS AND INTERESTS

- ♦ **Manufacturing & Fabrication:** CNC Machining, Manual Milling, Manual Lathe, Laser Cutting, 3D Printing, Drill Press Operations, Bandsaw Operations, Tapping & Threading (UNC/UNF Standards), Prototype Manufacturing
- ♦ **Software:** Fusion 360, AutoCAD, MATLAB, MotoCalc
- ♦ **Analysis:** Engineering Drawing Interpretation, Root Cause Analysis, Iterative Prototyping, Technical Documentation
- ♦ **Interpersonal Skills:** Positive Influence/Leadership in Team Environments, Conflict Resolution, Detail-Oriented

PROJECTS

Design Build Fly - Fixed Wing Aircraft

Fall 2025 - Present

- ♦ Collaborated within a multidisciplinary aeronautics subteam of 5-10 members over a 6-month competition cycle to design, develop, and manufacture a payload and mission-oriented fixed-wing aircraft to compete against 97 universities for the AIAA annual competition.
- ♦ Developed 2 static prototypes and 1 flight-capable prototype prior to final aircraft assembly to validate material selection, aerodynamic stability, manufacturability, and support a 15.05 lb gross-weight aircraft configuration carrying up to 38 passengers and 12 cargo units.
- ♦ Streamlined balsa and birch wood laser-cut fabrication in AutoCAD, and file organization for aircraft assemblies by incorporating 0.06in kerf compensation to improve press-fit dry assemblies.
- ♦ Assisted in propulsion subsystem testing and validation by evaluating 5 motor configurations using MotoCalc static simulations and experimental thrust stand testing to support a target cruise speed of 94.5 ft/s for the flight performance mission and 30.5-ft banner towing mission.
- ♦ Conducted 0–100% throttle sweeps to log thrust, current, voltage, power draw, and ESC temperature, contributing to the selection of the Scorpion SII-4025-520KV motor based on performance tradeoff analysis between cost, weight, efficiency, and power output.

CNC Handheld Projectile Launcher

Fall 2025

- ♦ Manufactured 6 mechanical components from detailed engineering drawings with tolerances as tight as ± 0.003 in, including threaded shafts, precision pins, chamfers, and counterbored 303 stainless steel, aluminum, and PMMA assemblies.
- ♦ Operated CNC mills, manual mills, and engine lathes to perform facing, drilling, threading, turning, chamfering, and tolerance-critical finishing operations while verifying dimensional accuracy against technical drawings.
- ♦ Applied machining and manufacturing principles, including tolerance analysis, thread standards (UNC/UNF), precision measurement, and CAD-to-fabrication workflows to produce functional multi-part mechanical assemblies.

Remote Controlled Projectile Robot - Team Project

Spring 2024

- ♦ Coordinated with a 4-person engineering team to design and fabricate a competition robot capable of collecting and launching 1-inch, 8-gram rubber projectiles into elevated scoring zones during a 60-second robotics competition.
- ♦ Modeled and refined drivetrain, intake, and flywheel launcher assemblies in Fusion 360 and AutoCAD, simplifying the robot architecture to reduce powered subsystems while improving manufacturability and assembly integration. 30.5-ft banner towing mission.
- ♦ Managed laser-cut fabrication workflows using Epilog Fusion LaserCam and applied kerf compensation/tolerance adjustments to improve press-fit assembly accuracy and reduce post-processing during integration.
- ♦ Performed experimental testing and root cause analysis on conflicts within flywheel launch and intake systems, identifying alignment instability and insufficient launch velocity as primary failure modes; implemented guide-path and geometry revisions to improve launch consistency and projectile control.
- ♦ Redesigned drivetrain traction system after identifying wheel slip during live testing, incorporating rubber-band traction surfaces to improve force transfer and robot mobility under payload conditions.

WORK EXPERIENCE

UCSD Housing and Dining Hospitality

April 2023 - Present

Student Lead Supervisor

- ♦ Trained and collaborated with team members in a fast-paced and high-pressure environment to develop efficient, capable workers, enhancing overall productivity, workflow, and team cohesion.
- ♦ Ensured strict compliance with safety and health protocols, maintaining corporate standards and professional work environments, and safeguarding operational integrity.